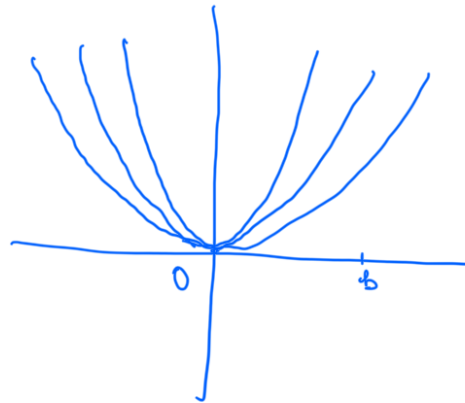
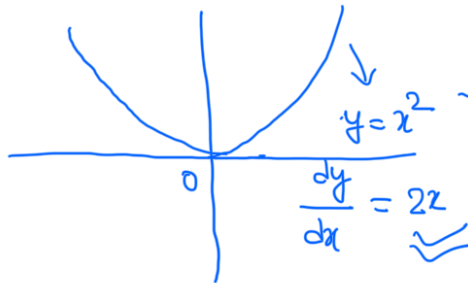


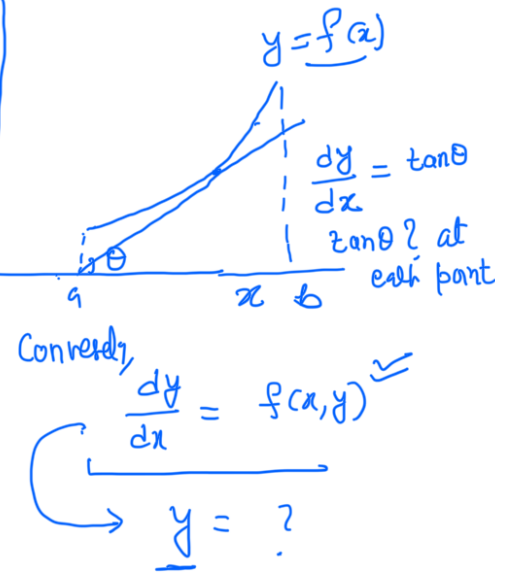
Lecture 1 (DE)

Book: Advanced Engg. Maths. E. Kreyszig

Diff. Eq. occurs naturally in physics, Engg. and so on. Velocity and acceleration being derivative, often give rise to diff Eq.



$$\begin{aligned} c=1, & \quad y=x^2 \\ c=2, & \quad y=2x^2 \\ & \vdots \end{aligned}$$



$$\begin{aligned} c > 0 \\ y &= c \cdot x^2 \Rightarrow c = \frac{y}{x^2}, \quad x \neq 0 \\ \frac{dy}{dx} &= 2cx \\ \frac{dy}{dx} &= \frac{2y}{x} \cdot x \\ \Rightarrow \frac{dy}{dx} &= \frac{2y}{x}, \quad x \neq 0 \\ \boxed{x \frac{dy}{dx} &= 2y} \end{aligned}$$

$$\text{Here, } \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{h}$$

Here, $y \rightarrow$ dependent variable, $x \rightarrow$ independent variable.

An equation involving derivatives of one or more dependent variable w.r. to one or more independent variables is called a differential Eq. (DE). If the no of independent variable is one then DE is an ODE

E.g. (i) $\frac{dy}{dx} = 2x$

(ii) $x \frac{dy}{dx} = 2y$

(iii) $\frac{d^2y}{dx^2} + mx = 0$

(iv) $\frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = e^x$

More precisely,

Definition: Let y be a function of x . An ordinary differential equation (ODE) is an equation containing one or more derivatives of an unknown y . Note that the eq. may contain y itself, and also x (including constants). In other words, an ODE is a relation between the derivatives, $y, y', y'', \dots, y^{(n)}$ and x : $F(x, y, y', y'', \dots, y^{(n)}) = 0$.

$y' = \frac{dy}{dx},$
 $y'' = \frac{d^2y}{dx^2},$
 $y^{(n)} = \frac{d^ny}{dx^n}.$

Order: The order of the highest derivative in the diff. eq is called the order of the diff. eq.

(i) 1 (ii) 1 (iii) 2 (iv) 2

(v) $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 + y = 0$, order 2

Degree: In general, the degree (positive integer exponent) of the diff eq. is the degree (exponent) of the highest order derivative in the diff. Eq (polynomial Eq. in all derivatives). Ex. (i) 1 (ii) 1 (iii) 1 (iv) 1 (v) 1.

(vi) $\left(\frac{d^3y}{dx^3}\right)^2 + \left(\frac{d^2y}{dx^2}\right)^3 = 0$, order 3, degree 2.

(vii) $\left(\frac{d^2y}{dx^2}\right)^{3/2} = y$, order 2, degree not defined.
 $\left(\frac{dy}{dx}\right)^3 = y^2$, degree 3.

Remark: If derivatives occur with real or fraction exponents the diff. Eq may not have a degree.

Formation: Exercise. (Eliminate all the arbitrary constants).

(i) $y = cx^2$, $x \frac{dy}{dx} = 2y$

(ii) $(x-a)^2 + (y-b)^2 = 1$, a, b - arbitrary constants.

(Order of the diff. Eq. = no. of the arbitrary constants).

Solution: A solution to an ODE $F(x, y, y', \dots, y^{(n)}) = 0$ is a function which satisfies the eq. in an open interval, i.e., n -times differentiable function y which satisfies the above eq. on some open interval (a, b) .

(i) $\frac{dy}{dx} = 0 \Rightarrow y = m$, m - arbitrary constant.
on whole real line $\mathbb{R}$

(ii) $\frac{dy}{dx} = m \Rightarrow y = mx + c$.
on $\mathbb{R}$

(iii) $\frac{dy}{dx} = 2x \Rightarrow y = x^2 + c$.
on $\mathbb{R}$

(iv) $y = \sqrt{9-x^2}$ is a solution
of $y \frac{dy}{dx} = -x$ on $(-3, 3)$.

The first graph shows a horizontal line $y = m$ on a coordinate system with axes x and y . The second graph shows a straight line $y = mx + c$ on a coordinate system with axes x and y . The line has a positive slope m and a y-intercept c . The angle θ is shown between the line and the positive x -axis, with $\tan \theta = m$. The third graph shows a parabola $y = x^2 + c$ on a coordinate system with axes x and y .

Some examples of naturally occurring diff. Eqs.

Ex.1 Radioactive decay:

A radioactive substance decomposes at a rate proportional to the amount present. Let $y(x)$ be the amount present at time x . Then

$$\frac{dy}{dx} = -ky,$$

$-k$ is called the decay constant (first order + first degree).

2) A falling object:

Suppose an object is falling under the force of gravity. The drag force which acts in the opposite direction to gravity can be assumed to be proportional to the velocity of the object. The drag force due to air resistance is cv^2 , where v is the velocity and c is the drag constant. By Newton's second law of motion

$$m \frac{dv}{dt} = mg - cv^2,$$

m - mass of the object, t - time.
(First order + first degree)

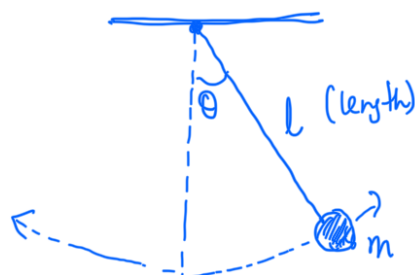
3) Simple harmonic motion:



$$m \frac{d^2 y}{dt^2} + ky = 0$$

(second order & first degree).

4) Oscillation of a simple pendulum:



$$\frac{d^2 \theta}{dt^2} + \frac{g}{l} \sin \theta = 0$$

(second order & first degree).

If the amplitude of angular displacement is small enough, the small angle approximation ($\sin \theta \sim \theta$) holds true, then the eq. of motion reduces to simple harmonic motion

$$\frac{d^2 \theta}{dt^2} + \frac{g}{L} \theta = 0.$$

